

WBN-FMS Calculation Note

Source: production model (congestion/predictor.py, bpr.py, queueing.py, physics.py)

Status: ACTIVE - this is what the app computes

Companion: planning_rules.md (the allocation rules this engine must obey)

1. Purpose and targets

The plan exists to meet the mine's sales targets with the trucks we have. Nothing below is a hidden factor. Each block is a named number. Allocation order is the same as planning_rules.md section 4.

Priority	Material	What we are trying to hit
P1	SAP	Saprolite to FeNi first (must-move). About 2,000 t/day per pit goes to POS as buffer; the rest goes direct to FeNi. IWIP reclaim must equal POS input.
P2	LIM-TOS	Limonite to the plant stockpile (HUAFEI / BSE). Fills only after SAP. BLB add-on +250 kt/month so four-month TOS LIM is 4.6 Mt, not the imported 3.6 Mt.
P3	LIM-LD	Leftover trucks haul limonite dump. Capacity is never clipped. Credited production stops at the supplied target (default 8 Mt on the horizon). The rest is named unused / excess.

A target is met when predicted tonnes sit on the day's target - not the inverse of required trucks. SAP and LIM-TOS must land. LIM-LD is judged at the bucket, with capacity and credited production shown as two labelled numbers.

2. The trips identity

A truck's productive loop plus the time that is not driving (breaks, dispatch, shift change) must fit inside the day. We attach that extra to the trip, not as a fraction of the clock. The old form was utilisation $U \times 1440 / \text{cycle}$, which claimed a truck cannot finish one trip once cycle exceeded $U \times 1440$.

$$T = 1440 / (\text{cycle} + \text{overhead})$$

T = trips per truck per day. 1440 min = two 12-hour shifts. cycle = productive minutes. overhead is fitted, not guessed.

Term	What it is	Where it is used
cycle	road_congested + ops + queue + bunching	Full cycle on the Congestion tab.
Plan cycle	road + ops only (queue and bunching dropped)	Plan and Allocate - owner: price the road.
overhead	Breaks, dispatch, shift change, fuel. Minutes per trip. Fitted so predict() at the median fleet equals the dispatch day-rate. Uncalibrated routes use the site median.	Every trips figure. Not a 0.85 availability multiplier.

$$WMT = T \times \text{payload} \times \text{trucks}$$

Payload is the contractor's measured tonnes per trip. This is Prediction. Achievable is a different engine (section 6).

3. Road delay - BPR and Akcelik

Theory: Bureau of Public Roads volume-delay (1964), the highway-planning standard. Extra time grows with how full the link is, to the fourth power. Above capacity a static BPR has no physics, so we switch to Akcelik (1991): oversaturation delay is a queue that grows with the shift length.

3.1 Below capacity ($v / c \leq 1$)

$$t = t_{\text{free}} \times (1 + 0.15 \times (v/c)^4)$$

alpha = 0.15, beta = 4. Literature defaults. Historical fleets here never gave a fitted pair. Quarter full: almost nothing. Near full: delay rises fast.

3.2 Above capacity ($v / c > 1$)

Delay grows with $(x - 1)$ and a square-root term, times the 12-hour period. $J = 2 \times \alpha$ so the two formulas meet at $v/c = 1$. Road time is still capped at 3 x free flow: trucks keep moving; they do not park for a shift.

3.3 Capacity and flow

$$c = \text{slowest posted speed} / 50 \text{ m gap (one loaded lane)}$$

$$v = \text{trucks on the section} / \text{cycle hours}$$

Owner speed-limit sheets. Separate loaded and empty lanes. Empty sits on the other carriageway. The slowest bin binds the whole section (averaging would overstate).

Section	Span	Min posted	Capacity
S1	TF - KR	30 km/h	600 trucks / hr
S2	KR - POS 12	30 km/h	600 trucks / hr
S3	POS 12 - KM 15	30 km/h	600 trucks / hr
S4	KM 15 - coast	20 km/h	400 trucks / hr

Stick routes split free time across S1-S4 by official limit times. Everyone on a section, including other companies as trucks/hr, shares that v . BLB is a spur that joins at chainage 2.45 km and counts on S4, not the whole mainline. UNI UNI / BIRI BIRI are out of scope.

Road load (v/c)	Status	Effect on the trip
under 0.7	GREEN	nearly none
0.7 - 1.0	YELLOW	minutes added
over 1.0	RED	queue grows all shift

4. Loader wait - closed queue

Theory: finite-source $M/M/c/N$, the machine-repair model (Koenigsberg 1958; Carmichael 1986; mining truck-shovel papers). The fleet is closed: the same N trucks circulate. They do not arrive from outside. Open Erlang-C would invent arrivals and explode when loaders saturate. We do not use that.

Each truck travels, then joins c loaders. Mean wait W_q comes from the birth-death balance of how many trucks are at the face. When demand outruns the loaders, throughput plateaus at

$$c \times (60 / \text{load_min}) \text{ trucks per hour}$$

No artificial cap. BPR's c is road geometry only. Folding loaders into road capacity made BLB look congested when three faces were full. Loader wait is charged once, here. Dump ceiling is reported separately, never mixed into BPR.

4.1 Bunching (arrival variance)

$$\text{extra} = 0.03 \times \text{cycle_sd} \times (N / N_{\text{ref}}) \text{ capped at } 3x$$

Trucks are not perfectly spaced. The term grows with fleet size. Same 3x wall as road time. Plan / Allocate omit it; Congestion shows it.

Cycle and flow depend on each other. The map is strictly decreasing, so there is one root. We find it by bisection, not a 50-step guess that used to sawtooth.

5. Rain, shared road, contractors

5.1 Rain - rolling resistance ratio

Mitschke and Spreeman (2023) haul-speed vs rolling-resistance curve. Wet road: rr rises 1 pp at ≥ 1 mm, or 2 pp at ≥ 10 mm. We do not replace the measured free-flow speed. We scale it by

$$\text{speed}(\text{rr_wet}) / \text{speed}(\text{rr_dry})$$

Dry is bit-identical (ratio = 1). Simulate tonnes stay weather-invariant; dwell can still move. Plan Step 1 WMT may move with rain.

5.2 Contractor baselines

Matched same-day history, both fleets on the same road, empirical-Bayes shrinkage, bounded 0.6-1.4. One road physics; contractors differ only in level (overhead re-anchored on the pooled cycle). Raw averages are fleet-size confounded (RIM ran about 140 trucks/day, SMA about 58) and are not used. On TF>POS 12, history has RIM slower than SMA. The old global 1.085x RIM factor is deleted.

Other companies enter as trucks per hour at their own tempo. They consume road. They do not add to our tonnes.

6. Two clocks

Clock	Engine	Used for	What it is
Prediction	Cycle model	Plan and Allocate	WMT = T x payload x trucks. Road + ops + overhead.
Achievable	Ticket engine (/api/simulate)	Run scenario	What weighbridge history says a fleet can move, with a loader clip.

We show both. We never average them. Hourly trucks-on-the-loaded-lane is a discrete-event picture of the finished plan. It distributes trucks already allocated. It does not re-price tonnes (congestion_clips_tonnes is always false). Occupancy is the loaded lane only. Colour is occupancy / trucks that fit at 50 m.

7. Hard rules the numbers must obey

These are the same walls as planning_rules.md. A calculation that breaks one of them is wrong, even if the formula is right.

- No 0.85 availability. Downtime is already inside trip time.
- Trucks keep their contractor. BLB is RIM only. KR is SMA only. SMA may work KR and TF; RIM may work BLB and TF. Never the reverse.
- IWIP POS-transit is its own fleet. Not taken from RIM or SMA. POS input = POS output.
- T11 is never a weighbridge. Identity is the matrix name, not the ticket number.
- We do not treat "the most we ever did" as "the most we can do" for road capacity.
- Fleet is conserved: used + free = pool. The allocator never invents trucks.